

Experiment 12 — WSL Implementation

nano producer_consumer.c

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#include <time.h>

#define NUM_PRODUCERS 2
#define NUM_CONSUMERS 2
#define ITEMS_PER_PRODUCER 10000

/* Change this value to test different buffer sizes */
#ifndef BUFFER_SIZE
#define BUFFER_SIZE 10
#endif

int buffer[BUFFER_SIZE];

int in = 0;
int out = 0;

sem_t empty;
sem_t full;

pthread_mutex_t mutex;

long produced_count = 0;
long consumed_count = 0;

/* Producer thread */
void *producer(void *arg)
{
    int id = *(int *)arg;

    for (int i = 0; i < ITEMS_PER_PRODUCER; i++)
```

```

{
    int item = id * ITEMS_PER_PRODUCER + i;

    /* Wait for an empty slot */
    sem_wait(&empty);

    /* Critical section */
    pthread_mutex_lock(&mutex);

    buffer[in] = item;
    in = (in + 1) % BUFFER_SIZE;
    produced_count++;

    pthread_mutex_unlock(&mutex);

    /* Signal that an item is available */
    sem_post(&full);
}

return NULL;
}

```

```

/* Consumer thread */
void *consumer(void *arg)
{
    int id = *(int *)arg;

    (void)id;

    for (int i = 0; i < ITEMS_PER_PRODUCER; i++)
    {
        /* Wait for an available item */
        sem_wait(&full);

        /* Critical section */
        pthread_mutex_lock(&mutex);

        int item = buffer[out];
        out = (out + 1) % BUFFER_SIZE;
        consumed_count++;
    }
}

```

```

pthread_mutex_unlock(&mutex);

/* Signal that a slot is empty */
sem_post(&empty);

/* Prevent compiler from optimizing away the read */
if (item < 0)
{
    printf("Invalid item\n");
}
}

return NULL;
}

int main()
{
    pthread_t producers[NUM_PRODUCERS];
    pthread_t consumers[NUM_CONSUMERS];

    int producer_ids[NUM_PRODUCERS];
    int consumer_ids[NUM_CONSUMERS];

    struct timespec start, end;

    printf("=====\n");
    printf("  PRODUCER-CONSUMER PROGRAM\n");
    printf("=====\n");

    printf("Buffer Size    : %d\n", BUFFER_SIZE);
    printf("Producers      : %d\n", NUM_PRODUCERS);
    printf("Consumers      : %d\n", NUM_CONSUMERS);
    printf("Items/Producer  : %d\n", ITEMS_PER_PRODUCER);
    printf("Total Items    : %d\n\n",
        NUM_PRODUCERS * ITEMS_PER_PRODUCER);

    /* Initialize semaphores */
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);

    /* Initialize mutex */

```

```
pthread_mutex_init(&mutex, NULL);

clock_gettime(CLOCK_MONOTONIC, &start);

/* Create producers */
for (int i = 0; i < NUM_PRODUCERS; i++)
{
    producer_ids[i] = i + 1;

    if (pthread_create(&producers[i], NULL,
                      producer, &producer_ids[i]) != 0)
    {
        perror("pthread_create");
        return 1;
    }
}

/* Create consumers */
for (int i = 0; i < NUM_CONSUMERS; i++)
{
    consumer_ids[i] = i + 1;

    if (pthread_create(&consumers[i], NULL,
                      consumer, &consumer_ids[i]) != 0)
    {
        perror("pthread_create");
        return 1;
    }
}

/* Wait for producers */
for (int i = 0; i < NUM_PRODUCERS; i++)
{
    pthread_join(producers[i], NULL);
}

/* Wait for consumers */
for (int i = 0; i < NUM_CONSUMERS; i++)
{
    pthread_join(consumers[i], NULL);
}
```

```
clock_gettime(CLOCK_MONOTONIC, &end);

double elapsed =
    (end.tv_sec - start.tv_sec) +
    (end.tv_nsec - start.tv_nsec) / 1e9;

double throughput = consumed_count / elapsed;

printf("Produced      : %ld\n", produced_count);
printf("Consumed      : %ld\n", consumed_count);
printf("Execution Time  : %.6f seconds\n", elapsed);
printf("Throughput     : %.2f items/second\n", throughput);

if (produced_count == consumed_count)
    printf("Synchronization : CORRECT\n");
else
    printf("Synchronization : ERROR\n");

printf("=====\n");

/* Destroy synchronization objects */
sem_destroy(&empty);
sem_destroy(&full);
pthread_mutex_destroy(&mutex);

return 0;
}
```

```
bharath@DESKTOP-6D34C2R: ~$ mkdir -p ~/ossp_exp12
cd ~/ossp_exp12
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ nano producer_consumer.c
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ gcc producer_consumer.c -o producer_consumer -pthread
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ ./producer_consumer
=====
PRODUCER-CONSUMER PROGRAM
=====
Buffer Size      : 10
Producers        : 2
Consumers        : 2
Items/Producer   : 10000
Total Items      : 20000

Produced         : 20000
Consumed         : 20000
Execution Time   : 0.083675 seconds
Throughput       : 239019.92 items/second
Synchronization : CORRECT
=====
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ gcc producer_consumer.c -o pc_1 -pthread -DBUFFER_SIZE=1
./pc_1
=====
PRODUCER-CONSUMER PROGRAM
=====
Buffer Size      : 1
Producers        : 2
Consumers        : 2
Items/Producer   : 10000
Total Items      : 20000

Produced         : 20000
Consumed         : 20000
Execution Time   : 0.770004 seconds
Throughput       : 25973.89 items/second
Synchronization : CORRECT
=====
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ gcc producer_consumer.c -o pc_5 -pthread -DBUFFER_SIZE=5
./pc_5
=====
PRODUCER-CONSUMER PROGRAM
=====
```

```
bharath@DESKTOP-6D34C2R: ~/ossp_exp12$ gcc producer_consumer.c -o pc_10 -pthread -DBUFFER_SIZE=10
./pc_10
=====
: 5
Producers      : 2
Consumers      : 2
Items/Producer : 10000
Total Items    : 20000

Produced       : 20000
Consumed       : 20000
Execution Time : 0.136286 seconds
Throughput     : 146750.66 items/second
Synchronization : CORRECT
=====
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ gcc producer_consumer.c -o pc_50 -pthread -DBUFFER_SIZE=50
./pc_50
=====
          PRODUCER-CONSUMER PROGRAM
=====
Buffer Size    : 50
Producers      : 2
Consumers      : 2
Items/Producer : 10000
Total Items    : 20000

Produced       : 20000
Consumed       : 20000
Execution Time : 0.066925 seconds
Throughput     : 298841.46 items/second
Synchronization : CORRECT
=====
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ gcc producer_consumer.c -o pc_10 -pthread -DBUFFER_SIZE=10
./pc_10
=====
          PRODUCER-CONSUMER PROGRAM
=====
Buffer Size    : 10
Producers      : 2
Consumers      : 2
Items/Producer : 10000
Total Items    : 20000

Produced       : 20000
Consumed       : 20000
Execution Time : 0.066925 seconds
Throughput     : 298841.46 items/second
Synchronization : CORRECT
=====
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ gcc producer_consumer.c -o pc_50 -pthread -DBUFFER_SIZE=50
./pc_50
=====
          PRODUCER-CONSUMER PROGRAM
=====
Buffer Size    : 50
Producers      : 2
Consumers      : 2
Items/Producer : 10000
Total Items    : 20000

Produced       : 20000
Consumed       : 20000
Execution Time : 0.013687 seconds
```

```
bharath@DESKTOP-6D34C2R: × + v - □ ×

=====
Buffer Size      : 100
Producers        : 2
Consumers        : 2
Items/Producer   : 10000
Total Items      : 20000

Produced         : 20000
Consumed         : 20000
Execution Time    : 0.016255 seconds
Throughput       : 1230416.84 items/second
Synchronization  : CORRECT
=====
bharath@DESKTOP-6D34C2R:~/ossp_exp12$ echo "===== BUFFER 1 ====="
/pc_1

echo ""
echo "===== BUFFER 5 ====="
/pc_5

echo ""
echo "===== BUFFER 10 ====="
/pc_10

echo ""
echo "===== BUFFER 50 ====="
/pc_50

echo ""
echo "===== BUFFER 100 ====="
/pc_100
===== BUFFER 1 =====
=====
          PRODUCER-CONSUMER PROGRAM
=====
Buffer Size      : 1
Producers        : 2
Consumers        : 2
Items/Producer   : 10000
Total Items      : 20000

Produced         : 20000
Consumed         : 20000
Execution Time    : 0.694128 seconds
Throughput       : 28813.12 items/second
Synchronization  : CORRECT
```

```
bharath@DESKTOP-6D34C2R: x + v

Producers      : 2
Consumers      : 2
Items/Producer : 10000
Total Items    : 20000

Produced       : 20000
Consumed       : 20000
Execution Time : 0.070739 seconds
Throughput     : 282728.59 items/second
Synchronization : CORRECT
=====

===== BUFFER 50 =====
=====
          PRODUCER-CONSUMER PROGRAM
=====
Buffer Size    : 50
Producers      : 2
Consumers      : 2
Items/Producer : 10000
Total Items    : 20000

Produced       : 20000
Consumed       : 20000
Execution Time : 0.020324 seconds
Throughput     : 984065.18 items/second
Synchronization : CORRECT
=====

===== BUFFER 100 =====
=====
          PRODUCER-CONSUMER PROGRAM
=====
Buffer Size    : 100
Producers      : 2
Consumers      : 2
Items/Producer : 10000
Total Items    : 20000

Produced       : 20000
Consumed       : 20000
Execution Time : 0.022134 seconds
Throughput     : 903575.65 items/second
Synchronization : CORRECT
=====
bharath@DESKTOP-6D34C2R:~/ossp_exp12$
```